A11 Box - Wire List

United Control Box, wiring rev 0.7 · 2026-10-27
One row per wire, in build-guide order · lengths from the 3D duct routing

104 single wires, 95.2 m to cut (+ 4 multi-core cables, last table). **Cut** = routed + 10 % + 15 cm, rounded up to 5 cm; door wires include 0.30 m for the hinge loop. **Marking**: the wire number on a sleeve at **both ends**. All 1 mm² wire is yellow, so its function colour goes on as a **band** (coloured sleeve or tape) next to the number at both ends: black = L / DC power, blue = N, red = 230 V AC control (KM1 coil), dark blue = +24 V / DC control / signals, blue + white = 0 V. Never use plain yellow for earth: every PE wire is G/Y. **2×** under Ends = twin ferrule shared with the wire named.

Stock check (cut lengths against the wire you have)

Wire (reel)	Pieces	To cut (m)	Have (m)	Left / short (m)	Status
1 mm² yellow	77	69.20	90	+20.80	OK
2.5 mm² G/Y	6	7.95	4	-3.95	**SHORT 3.95 m**
2.5 mm² black	2	2.50	4	+1.50	OK
2.5 mm² blue	2	2.80	4	+1.20	OK
2.5 mm² brown	2	2.10	4	+1.90	OK
4 mm² G/Y	5	4.10	5	+0.90	OK
4 mm² black	7	3.90	5	+1.10	OK
4 mm² blue	2	1.25	5	+3.75	OK
6 mm² braided strap	1	1.40			to buy (parts list)
4 mm² brown	0	0.00	5	+5.00	spare

Ferrules needed: single 1 mm² ×77, single 2.5 mm² ×10, single 4 mm² ×17, twin 2 × 1 mm² ×19, twin 2 × 4 mm² ×1.

Open points

- **2.5 mm² G/Y: short by 3.95 m**, buy 5 m (or 1.5 mm² G/Y for the PSU / outlet PE wires 145, 146, 148, 149).
- **48 V drive feeds 150-155 in 1 mm²** (10.4 A in a duct) are below the 15 A of the 48 V PSU: wiring rule R3 fails for 150 / 152 / 154. Fix with one fuse per drive (6.3 A, holder on the DIN rail) or wire them in 1.5-2.5 mm² (about 6 m). The drives draw far less in normal use; the rule covers a short circuit.
- **Braking resistor leads 128 / 129** are 1 mm² PVC (owner's choice): keep them clear of the tube, which gets very hot.

1 · Protective earth - 12 wires, 13.45 m

	No.	From	To	Wire	Marking (both ends)	Cut m	Ends (from / to)	Function
☐	**138**	Earth bar 1	Earth bar 2	4 mm² G/Y	**138**	**0.60**	ring lug M6 ring lug M6	link (both bars sit on insulators)
☐	**139**	Enclosure body (earth stud)	Earth bar 1	4 mm² G/Y	**139**	**0.80**	ring lug M6 ring lug M6	NEW: the body had no PE conductor
☐	**140**	Mains (gland) PE	Earth bar 1	4 mm² G/Y	**140**	**0.65**	supply cable core from the gland ring lug M6	
☐	**141**	Backplate	Earth bar 1	4 mm² G/Y	**141**	**0.45**	ring lug M6 ring lug M6	
☐	**142**	Door strap	Earth bar 1	6 mm² braided strap	**142**	**1.40**	ring lug M6 ring lug M6	door wire (hinge loop)
☐	**143**	EMI filter GND	Earth bar 1	4 mm² G/Y	**143**	**1.60**	ferrule 4 mm² ring lug M6	
☐	**144**	VFD PEc	Earth bar 1	2.5 mm² G/Y	**144**	**1.15**	ferrule 2.5 mm² ring lug M6	
☐	**145**	PSU 48V PE	Earth bar 2	2.5 mm² G/Y	**145**	**1.30**	insulated fork lug ring lug M6	
☐	**146**	PSU 24V PE	Earth bar 2	2.5 mm² G/Y	**146**	**1.15**	insulated fork lug ring lug M6	
☐	**147**	DIN-rail earth stud	Earth bar 2	2.5 mm² G/Y	**147**	**1.65**	ring lug M6 ring lug M6	
☐	**148**	PSU 24V -V2	Earth bar 2	2.5 mm² G/Y	**148**	**1.20**	2 fork lugs under one screw with 169 ring lug M6	PELV bond: the only 0 V to PE bond
☐	**149**	Schuko PE	Earth bar 2	2.5 mm² G/Y	**149**	**1.50**	ferrule 2.5 mm² ring lug M6	NEW: missing in rev 0.5 (pump was unearthed)

2 · 230 V power and VFD - 24 wires, 19.35 m

	No.	From	To	Wire	Marking (both ends)	Cut m	Ends (from / to)	Function
☐	**101**	Mains (gland) L	Main CB 1	2.5 mm² black	**101**	**1.50**	supply cable core from the gland ferrule 2.5 mm²	Mains L
☐	**102**	Mains (gland) N	Main CB 3	2.5 mm² blue	**102**	**1.55**	supply cable core from the gland ferrule 2.5 mm²	Mains N
☐	**103**	Main CB 2	EMI filter LIN	4 mm² black	**103**	**0.30**	ferrule 4 mm² ferrule 4 mm²	
☐	**104**	Main CB 4	EMI filter NIN	4 mm² blue	**104**	**0.35**	ferrule 4 mm² ferrule 4 mm²	
☐	**105**	EMI filter LOUT	Contactor 1L1	4 mm² black	**105**	**0.45**	2× 4 mm² with 111 ferrule 4 mm²	Unswitched L
☐	**106**	EMI filter NOUT	N busbar XN	4 mm² blue	**106**	**0.90**	ferrule 4 mm² ferrule 4 mm² (own point on the distribution block)	N in
☐	**109**	Contactor 2T1	L busbar XL	4 mm² black	**109**	**0.75**	ferrule 4 mm² ferrule 4 mm² (own point on the distribution block)	Switched L in
☐	**111**	EMI filter LOUT	CB-24V 1	4 mm² black	**111**	**0.55**	2× 4 mm² with 105 ferrule 4 mm²	24 V PSU L (unswitched)

	No.	From	To	Wire	Marking (both ends)	Cut m	Ends (from / to)	Function
☐	**112**	CB-24V 2	PSU 24V L	1 mm² yellow	**112** + **black** band	**1.00**	2× 1 mm² with 130 insulated fork lug	
☐	**113**	N busbar XN	PSU 24V N	1 mm² yellow	**113** + **blue** band	**0.85**	ferrule 1 mm² (own point on the distribution block) insulated fork lug	
☐	**114**	L busbar XL	CB-VFD 1	4 mm² black	**114**	**0.60**	ferrule 4 mm² (own point on the distribution block) ferrule 4 mm²	Switched L
☐	**115**	CB-VFD 2	VFD L	2.5 mm² black	**115**	**1.00**	ferrule 2.5 mm² ferrule 2.5 mm²	
☐	**116**	N busbar XN	VFD N	2.5 mm² blue	**116**	**1.25**	ferrule 2.5 mm² (own point on the distribution block) ferrule 2.5 mm²	
☐	**117**	L busbar XL	CB-48V 1	4 mm² black	**117**	**0.60**	ferrule 4 mm² (own point on the distribution block) ferrule 4 mm²	Switched L
☐	**118**	CB-48V 2	PSU 48V L	1 mm² yellow	**118** + **black** band	**1.15**	ferrule 1 mm² insulated fork lug	
☐	**119**	N busbar XN	PSU 48V N	1 mm² yellow	**119** + **blue** band	**0.95**	ferrule 1 mm² (own point on the distribution block) insulated fork lug	
☐	**120**	L busbar XL	CB-PUMP 1	4 mm² black	**120**	**0.65**	ferrule 4 mm² (own point on the distribution block) ferrule 4 mm²	Switched L
☐	**121**	CB-PUMP 2	Schuko L	1 mm² yellow	**121** + **black** band	**0.40**	ferrule 1 mm² ferrule 1 mm²	Pump
☐	**122**	N busbar XN	Schuko N	1 mm² yellow	**122** + **blue** band	**0.45**	ferrule 1 mm² (own point on the distribution block) ferrule 1 mm²	Pump
☐	**128**	VFD P+	Braking resistor T1	1 mm² yellow	**128** + **black** band	**1.25**	ferrule 1 mm² ring lug (resistor stud)	
☐	**129**	VFD PB	Braking resistor T2	1 mm² yellow	**129** + **black** band	**1.00**	ferrule 1 mm² ring lug (resistor stud)	
☐	**130**	CB-24V 2	K1 41 (NEMA 12)	1 mm² yellow	**130** + **red** band	**0.85**	2× 1 mm² with 112 ferrule 1 mm²	Coil supply L (protected by QF4 C6)
☐	**131**	K1 44 (NEMA 8)	Contactor A2	1 mm² yellow	**131** + **red** band	**0.35**	ferrule 1 mm² ferrule 1 mm²	Switched coil L
☐	**132**	N busbar XN	Contactor A1	1 mm² yellow	**132** + **blue** band	**0.65**	ferrule 1 mm² (own point on the distribution block) ferrule 1 mm²	Coil N

3 · 48 V drive supply - 6 wires, 6.10 m

	No.	From	To	Wire	Marking (both ends)	Cut m	Ends (from / to)	Function
☐	**150**	PSU 48V +V1	DMA860H-X AC1	1 mm² yellow	**150** + **black** band	**1.00**	insulated fork lug ferrule 1 mm²	48 V
☐	**151**	PSU 48V -V1	DMA860H-X AC2	1 mm² yellow	**151** + **black** band	**0.95**	insulated fork lug ferrule 1 mm²	
☐	**152**	PSU 48V +V2	DMA860H-Y AC1	1 mm² yellow	**152** + **black** band	**1.10**	insulated fork lug ferrule 1 mm²	
☐	**153**	PSU 48V -V2	DMA860H-Y AC2	1 mm² yellow	**153** + **black** band	**1.05**	insulated fork lug ferrule 1 mm²	
☐	**154**	PSU 48V +V3	DMA860H-Z AC1	1 mm² yellow	**154** + **black** band	**1.00**	insulated fork lug ferrule 1 mm²	
☐	**155**	PSU 48V -V3	DMA860H-Z AC2	1 mm² yellow	**155** + **black** band	**1.00**	insulated fork lug ferrule 1 mm²	

4 · 24 V control - 32 wires, 28.45 m

	No.	From	To	Wire	Marking (both ends)	Cut m	Ends (from / to)	Function
☐	**162**	PSU 24V +V1	F1 in	2.5 mm² brown	**162** + **dark blue** band	**1.10**	insulated fork lug ferrule 2.5 mm²	Control fuse (short, 2.5 mm²: unfused PSU side)
☐	**164**	PSU 24V +V2	F2 in	2.5 mm² brown	**164** + **dark blue** band	**1.00**	insulated fork lug ferrule 2.5 mm²	Signal fuse (short, 2.5 mm²: unfused PSU side)
☐	**166**	F2 out	A11 DC24V	1 mm² yellow	**166** + **dark blue** band	**1.40**	2× 1 mm² with 168 ferrule 1 mm²	
☐	**167**	PSU 24V -V1	A11 DCGND	1 mm² yellow	**167** + **blue/white** band	**0.95**	2 fork lugs under one screw with 207 2× 1 mm² with 171	
☐	**168**	F2 out	A11 24V	1 mm² yellow	**168** + **dark blue** band	**1.40**	2× 1 mm² with 166 2× 1 mm² with 340	Sensor/input supply
☐	**169**	PSU 24V -V2	A11 GND	1 mm² yellow	**169** + **blue/white** band	**1.00**	2 fork lugs under one screw with 148 2× 1 mm² with 337	
☐	**170**	Contactor 43	Fan +	1 mm² yellow	**170** + **dark blue** band	**1.35**	2× 1 mm² with 220 ferrule 1 mm²	NEW: fan behind F1 (was unfused on TB24+)
☐	**171**	A11 DCGND	Fan -	1 mm² yellow	**171** + **blue/white** band	**0.85**	2× 1 mm² with 167 ferrule 1 mm²	
☐	**201**	F1 out	E-stop 1	1 mm² yellow	**201** + **dark blue** band	**1.90**	2× 1 mm² with 220 2× 1 mm² with 223	Chain; door wire (hinge loop)
☐	**202**	E-stop 2	OFF 1	1 mm² yellow	**202** + **dark blue** band	**0.35**	ferrule 1 mm² ferrule 1 mm²	Chain; door wire (hinge loop)
☐	**203**	OFF 2	ON 1	1 mm² yellow	**203** + **dark blue** band	**0.25**	2× 1 mm² with 205 ferrule 1 mm²	Chain; door wire (hinge loop)
☐	**204**	ON 2	K1 A1 (NEMA 13)	1 mm² yellow	**204** + **dark blue** band	**1.70**	ferrule 1 mm² 2× 1 mm² with 206	Start; door wire (hinge loop)
☐	**205**	OFF 2	K1 11 (NEMA 9)	1 mm² yellow	**205** + **dark blue** band	**1.85**	2× 1 mm² with 203 ferrule 1 mm²	Latch; door wire (hinge loop)

	No.	From	To	Wire	Marking (both ends)	Cut m	Ends (from / to)	Function
☐	**206**	K1 14 (NEMA 5)	K1 A1 (NEMA 13)	1 mm² yellow	**206** + **dark blue** band	**0.80**	ferrule 1 mm² 2× 1 mm² with 204	Latch
☐	**207**	PSU 24V -V1	K1 A2 (NEMA 14)	1 mm² yellow	**207** + **blue/white** band	**0.95**	2 fork lugs under one screw with 167 ferrule 1 mm²	Coil return
☐	**220**	F1 out	Contactor 43	1 mm² yellow	**220** + **dark blue** band	**1.05**	2× 1 mm² with 201 2× 1 mm² with 170	Yellow lamp supply
☐	**221**	Contactor 44	Yellow lamp +	1 mm² yellow	**221** + **dark blue** band	**1.70**	ferrule 1 mm² ferrule 1 mm²	door wire (hinge loop)
☐	**222**	K1 A2 (2nd screw) (NEMA 14)	Yellow lamp -	1 mm² yellow	**222** + **blue/white** band	**1.75**	ferrule 1 mm² ferrule 1 mm²	0 V to the door; door wire (hinge loop)
☐	**223**	E-stop 1	Green lamp +	1 mm² yellow	**223** + **dark blue** band	**0.50**	2× 1 mm² with 201 2× 1 mm² with 225	+24 V CONTROL (door loop); door wire (hinge loop)
☐	**225**	Green lamp +	Red lamp +	1 mm² yellow	**225** + **dark blue** band	**0.30**	2× 1 mm² with 223 ferrule 1 mm²	door wire (hinge loop)
☐	**316**	A11 OGND	DMA860H-X ALM-	1 mm² yellow	**316** + **blue/white** band	**0.75**	2× 1 mm² with 326 2× 1 mm² with 318	
☐	**318**	DMA860H-X ALM-	DMA860H-Y ALM-	1 mm² yellow	**318** + **blue/white** band	**0.55**	2× 1 mm² with 316 2× 1 mm² with 320	
☐	**320**	DMA860H-Y ALM-	DMA860H-Z ALM-	1 mm² yellow	**320** + **blue/white** band	**0.50**	2× 1 mm² with 318 ferrule 1 mm²	
☐	**326**	A11 OGND	VFD COM	1 mm² yellow	**326** + **blue/white** band	**0.80**	2× 1 mm² with 316 ferrule 1 mm²	Output common
☐	**337**	A11 GND	X-LIM 3	1 mm² yellow	**337** + **blue/white** band	**0.75**	2× 1 mm² with 169 solder cup (GX16 connector)	Sensor 0 V (BU)
☐	**338**	X-LIM 3	Y-LIM 3	1 mm² yellow	**338** + **blue/white** band	**0.40**	solder cup (GX16 connector) solder cup (GX16 connector)	Sensor 0 V (loop)
☐	**339**	Y-LIM 3	Z-LIM 3	1 mm² yellow	**339** + **blue/white** band	**0.40**	solder cup (GX16 connector) solder cup (GX16 connector)	Sensor 0 V (loop)
☐	**340**	A11 24V	X-LIM 4	1 mm² yellow	**340** + **dark blue** band	**0.75**	2× 1 mm² with 168 solder cup (GX16 connector)	Sensor +24 V (BN)
☐	**341**	X-LIM 4	Y-LIM 4	1 mm² yellow	**341** + **dark blue** band	**0.40**	solder cup (GX16 connector) solder cup (GX16 connector)	Sensor +24 V (loop)
☐	**342**	Y-LIM 4	Z-LIM 4	1 mm² yellow	**342** + **dark blue** band	**0.40**	solder cup (GX16 connector) solder cup (GX16 connector)	Sensor +24 V (loop)
☐	**346**	Z-LIM 3	Tool setter 2	1 mm² yellow	**346** + **blue/white** band	**0.40**	solder cup (GX16 connector) solder cup (GX16 connector)	0 V (loop)
☐	**348**	Tool setter 2	Tool setter 4	1 mm² yellow	**348** + **blue/white** band	**0.20**	solder cup (GX16 connector) solder cup (GX16 connector)	0 V to the over-travel switch (jumper in the connector)

5 · Signals - 30 wires, 27.85 m

	No.	From	To	Wire	Marking (both ends)	Cut m	Ends (from / to)	Function
☐	**210**	E-stop 4	A11 X7	1 mm² yellow	**210** + **dark blue** band	**2.00**	ferrule 1 mm² ferrule 1 mm²	E-stop signal to the controller; door wire (hinge loop)
☐	**224**	Green lamp -	A11 Y6	1 mm² yellow	**224** + **dark blue** band	**2.00**	ferrule 1 mm² ferrule 1 mm²	Work; door wire (hinge loop)
☐	**226**	Red lamp -	A11 Y5	1 mm² yellow	**226** + **dark blue** band	**2.05**	ferrule 1 mm² ferrule 1 mm²	Alarm; door wire (hinge loop)
☐	**301**	A11 X_5V	DMA860H-X PUL+	1 mm² yellow, twisted	**301** + **dark blue** band	**0.85**	ferrule 1 mm² 2× 1 mm² with 304	
☐	**302**	A11 X_PUL	DMA860H-X PUL-	1 mm² yellow, twisted	**302** + **dark blue** band	**0.85**	ferrule 1 mm² ferrule 1 mm²	
☐	**303**	A11 X_DIR	DMA860H-X DIR-	1 mm² yellow, twisted	**303** + **dark blue** band	**0.95**	ferrule 1 mm² ferrule 1 mm²	
☐	**304**	DMA860H-X PUL+	DMA860H-X DIR+	1 mm² yellow	**304** + **dark blue** band	**0.20**	2× 1 mm² with 301 ferrule 1 mm²	Jumper
☐	**305**	A11 Y_5V	DMA860H-Y PUL+	1 mm² yellow, twisted	**305** + **dark blue** band	**1.05**	ferrule 1 mm² 2× 1 mm² with 308	
☐	**306**	A11 Y_PUL	DMA860H-Y PUL-	1 mm² yellow, twisted	**306** + **dark blue** band	**1.05**	ferrule 1 mm² ferrule 1 mm²	
☐	**307**	A11 Y_DIR	DMA860H-Y DIR-	1 mm² yellow, twisted	**307** + **dark blue** band	**1.05**	ferrule 1 mm² ferrule 1 mm²	
☐	**308**	DMA860H-Y PUL+	DMA860H-Y DIR+	1 mm² yellow	**308** + **dark blue** band	**0.20**	2× 1 mm² with 305 ferrule 1 mm²	Jumper
☐	**309**	A11 Z_5V	DMA860H-Z PUL+	1 mm² yellow, twisted	**309** + **dark blue** band	**1.15**	ferrule 1 mm² 2× 1 mm² with 312	
☐	**310**	A11 Z_PUL	DMA860H-Z PUL-	1 mm² yellow, twisted	**310** + **dark blue** band	**1.25**	ferrule 1 mm² ferrule 1 mm²	
☐	**311**	A11 Z_DIR	DMA860H-Z DIR-	1 mm² yellow, twisted	**311** + **dark blue** band	**1.25**	ferrule 1 mm² ferrule 1 mm²	
☐	**312**	DMA860H-Z PUL+	DMA860H-Z DIR+	1 mm² yellow	**312** + **dark blue** band	**0.20**	2× 1 mm² with 309 ferrule 1 mm²	Jumper
☐	**315**	DMA860H-X ALM+	A11 X5	1 mm² yellow	**315** + **dark blue** band	**0.75**	2× 1 mm² with 317 ferrule 1 mm²	Drive alarm (wire-OR)

	No.	From	To	Wire	Marking (both ends)	Cut m	Ends (from / to)	Function
☐	**317**	DMA860H-Y ALM+	DMA860H-X ALM+	1 mm² yellow	**317** + **dark blue** band	**0.55**	2× 1 mm² with 319 2× 1 mm² with 315	NEW: X5 had 3 conductors
☐	**319**	DMA860H-Z ALM+	DMA860H-Y ALM+	1 mm² yellow	**319** + **dark blue** band	**0.55**	ferrule 1 mm² 2× 1 mm² with 317	
☐	**325**	A11 Y1	VFD X1	1 mm² yellow	**325** + **dark blue** band	**0.70**	ferrule 1 mm² ferrule 1 mm²	Spindle run (forward)
☐	**327**	A11 Y2	VFD X2	1 mm² yellow	**327** + **dark blue** band	**0.70**	ferrule 1 mm² ferrule 1 mm²	Multi-speed bit 1
☐	**328**	A11 Y3	VFD X3	1 mm² yellow	**328** + **dark blue** band	**0.75**	ferrule 1 mm² ferrule 1 mm²	Multi-speed bit 2
☐	**329**	A11 Y4	VFD X4	1 mm² yellow	**329** + **dark blue** band	**0.75**	ferrule 1 mm² ferrule 1 mm²	Multi-speed bit 3
☐	**330**	X-LIM 1	A11 X1	1 mm² yellow	**330** + **dark blue** band	**0.70**	solder cup (GX16 connector) ferrule 1 mm²	X home (X−)
☐	**331**	Y-LIM 1	A11 X2	1 mm² yellow	**331** + **dark blue** band	**0.75**	solder cup (GX16 connector) ferrule 1 mm²	Y home (Y−)
☐	**332**	Z-LIM 2	A11 X3	1 mm² yellow	**332** + **dark blue** band	**0.80**	solder cup (GX16 connector) ferrule 1 mm²	Z home (Z+)
☐	**334**	X-LIM 2	Y-LIM 2	1 mm² yellow	**334** + **dark blue** band	**0.40**	solder cup (GX16 connector) solder cup (GX16 connector)	Y+ limit (loop)
☐	**335**	Y-LIM 2	Z-LIM 1	1 mm² yellow	**335** + **dark blue** band	**0.40**	solder cup (GX16 connector) solder cup (GX16 connector)	Z− limit (loop)
☐	**336**	X-LIM 2	A11 X6	1 mm² yellow	**336** + **dark blue** band	**0.75**	solder cup (GX16 connector) ferrule 1 mm²	Hard limit (wire-OR)
☐	**345**	Tool setter 1	A11 X4	1 mm² yellow	**345** + **dark blue** band	**0.85**	solder cup (GX16 connector) ferrule 1 mm²	Tool-setter touch
☐	**347**	Tool setter 3	E-stop 3	1 mm² yellow	**347** + **dark blue** band	**2.35**	solder cup (GX16 connector) ferrule 1 mm²	Tool-setter over-travel (NC) in series with the E-stop NC2 → A11 X7 (door loop); door wire (hinge loop)

Multi-core cables (cut once per cable, to the longest core)

	Cable	Cores (no.: from → to)	Cut m
☐	Motor cable X: LiYCY 4 × 1.0 shielded	DMA860H-X A+ → XM-X 1 DMA860H-X A- → XM-X 2 DMA860H-X B+ → XM-X 3 DMA860H-X B- → XM-X 4	**1.15**
☐	Motor cable Y: LiYCY 4 × 1.0 shielded	DMA860H-Y A+ → XM-Y 1 DMA860H-Y A- → XM-Y 2 DMA860H-Y B+ → XM-Y 3 DMA860H-Y B- → XM-Y 4	**1.00**
☐	Motor cable Z: LiYCY 4 × 1.0 shielded	DMA860H-Z A+ → XM-Z 1 DMA860H-Z A- → XM-Z 2 DMA860H-Z B+ → XM-Z 3 DMA860H-Z B- → XM-Z 4	**0.95**
☐	Spindle cable W1: 4G1.5 shielded	123: VFD U → HDC-HA 1 124: VFD V → HDC-HA 2 125: VFD W → HDC-HA 3 126: VFD PE → HDC-HA PE 127: HDC-HA SH → VFD PEb	**1.05**

Not listed: 12 conductors outside the box (motor and sensor cables to the machine).
Source: Verification/A11_Wire_Lengths_rev07.csv (player -a11lengths); regenerate with make_wire_list.py.